PRODUCT REQUIREMENTS DOCUMENT

AI Platform Growth Product Analytics Project

User Activation, Retention and Growth across AI Platforms

Claude · ChatGPT · Gemini · Grok

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1. Executive Summary

This Product Requirements Document defines the analytical framework for the AI Platform Growth Product Analytics Project — a comprehensive product analytics case study comparing user engagement, retention, and monetisation behavior across the four leading AI assistant products: Claude (Anthropic), ChatGPT (OpenAI), Gemini (Google), and Grok (xAI).

This project simulates the work of a product analyst embedded within a product or growth team, tasked with generating actionable insights that inform product roadmap decisions. It demonstrates end-to-end analytical capability — from problem framing and metric definition through SQL analysis, event tracking, dashboard creation, and stakeholder communication.

WHY THIS MATTERS	The AI assistant market is one of the fastest-growing product categories in history. Understanding what drives user activation, retention, and conversion is critical for any product team competing in this space. This analysis surfaces the behavioral patterns that separate sticky products from leaky ones.

2. Problem Statement

2.1 The Business Problem

AI assistant products share a common challenge: high acquisition but poor retention. Users sign up in large numbers — driven by hype, media coverage, or employer adoption — but a significant portion never becomes habitually engaged. The pattern is predictable:

- Day 0: User signs up and sends their first message
- Day 1–3: Usage spikes as the novelty effect kicks in
- Day 7: A large drop-off occurs — many users simply stop returning
- Day 30: Only a small fraction remain active

This is not just a retention problem — it is a revenue problem. Subscription conversion, premium tier upgrades, and long-term lifetime value all depend on users building a habit around the product.

2.2 Analytical Questions We Are Solving

This project is structured around five core analytical questions that a real product team would need answered:

#	Analytical Question	SQL / Tool	Priority	Phase

Q1	Where exactly do users drop off during the onboarding funnel, and which step loses the most users?	SQL funnel analysis + Mixpanel	Critical	Phase 3
Q2	Which AI product has the strongest 7-day and 30-day retention, and what behavioral signals predict long-term retention?	SQL cohort + Mixpanel retention	Critical	Phase 3–4
Q3	Did the 'suggested prompts' onboarding feature increase Day-1 activation? (A/B test analysis)	SQL + Statistics	High	Phase 3
Q4	Which product features (code generation, image input, memory, plugins) are most associated with retained users?	SQL aggregation + Mixpanel	High	Phase 3–4
Q5	What is the revenue impact of improving Day-7 retention by 5 percentage points?	SQL + Excel modeling	Medium	Phase 6

2.3 Scope

- **Time period: 6 months of simulated data (Nov 2025 – Apr 2026)** In Scope:
- **Products: Claude, ChatGPT, Gemini, Grok** In Scope:
- **User segments: Free tier, Pro tier, Team/Enterprise tier** In Scope:
- **Platforms: Web, iOS, Android** In Scope:
- **Real user data (all data is synthetic and generated for portfolio purposes)** Out of Scope:
- **API usage patterns (consumer product only)** Out of Scope:

3. User Personas & Segments

Understanding who our users are is essential before defining what we measure. This project tracks four distinct user segments, each with different behaviors and retention expectations.

Persona	Profile	Primary Use Case	Retention Risk
Power User	Daily active, Pro/Team tier, 10+ sessions/week	Code generation, research, writing	Low — highly habituated, but may switch if a rival releases a killer feature
Casual Explorer	Weekly active, Free tier, 2–4 sessions/week	General Q&A, curiosity-driven queries	Medium — engagement is shallow; easily disrupted by friction or novelty elsewhere
Task Triager	Sporadic active, Free tier, used for specific tasks	One-off tasks: summarise a doc, draft an email	High — no habit formed; likely to churn after first use case is resolved
Churned User	Not active in 30+ days, signed up but disengaged	Unknown — stopped before habituating	Already churned — target for re-engagement campaign analysis

4. Metrics Framework

4.1 The North Star Metric

NORTH STAR Weekly Active Users who complete 5 or more meaningful interactions (WAU-5). A 'meaningful interaction' is defined as a conversation thread of 2+ message turns — not a single-send query. This measures genuine, habitual engagement rather than surface-level logins.

Why WAU-5 and not DAU or MAU? DAU overcounts — a user who logs in to check a single thing is not "engaged". MAU undercounts — it includes users who are barely active. WAU-5 captures the habit loop: users returning multiple times within a week and having substantive conversations.

4.2 Primary Metrics (L1)

These are the metrics the product team tracks weekly in executive dashboards.

Metric	Definition	Target	Owner
WAU-5	Weekly active users with 5+ conversation turns	> 35% of registered users	Product
Day-7 Retention	% of new users who return on Day 7	> 40%	Growth
Day-30 Retention	% of new users still active on Day 30	> 20%	Growth
Free-to-Pro Conversion	% of free users who upgrade to Pro within 30 days	> 8%	Revenue
DAU/MAU Ratio	Engagement density — how often monthly users return	> 0.30	Product

4.3 Secondary Metrics (L2)

These metrics support diagnosis — they explain why L1 metrics are moving.

Metric	Definition	Target	Owner
Activation Rate	% of sign-ups who send their first message within 24h	> 70%	Onboarding
Feature Adoption Rate	% of active users who use at least 1 advanced feature	> 30%	Product

Average Session Length	Mean time spent per session (minutes)	> 8 min	Engagement
Messages per Session	Mean conversation turns per session	> 4 turns	Engagement
Churn Rate (30-day)	% of previously active users with 0 sessions in 30 days	< 25%	Retention

4.4 Guardrail Metrics

These must not degrade — they protect user trust and product health.

- Response satisfaction rate (thumbs up/down) must stay above 75%
- Error / failure rate per session must stay below 2%
- Support ticket volume must not increase by more than 10% week-over-week

5. Event Tracking Plan (Mixpanel)

This section defines the Mixpanel event taxonomy — every user action we will track and the properties attached to each event. A clean, well-defined tracking plan is foundational to reliable analytics.

5.1 Core Events

Event Name	Trigger	Key Properties
account_created	User completes sign-up form	user_id, product, signup_source, plan_type, country, platform
onboarding_step_completed	User completes a step in the onboarding flow	user_id, step_name, step_number, time_on_step_seconds, ab_variant
session_started	User opens the app / loads a new session	user_id, product, platform, session_id, days_since_signup
message_sent	User submits a message in a conversation	user_id, session_id, message_length, turn_number, feature_used
feature_used	User triggers an advanced feature	user_id, feature_name [code/image/memory/plugin], session_id
subscription_upgraded	User upgrades from Free to Pro or Team	user_id, from_plan, to_plan, revenue_usd, days_since_signup
session_ended	User closes the app or session times out	user_id, session_id, session_duration_seconds, messages_in_session

churn_signal	User has 0 sessions for 14 consecutive days	user_id, last_session_date, days_inactive, last_feature_used

5.2 A/B Test Event

For the 'Suggested Prompts' experiment, we add a specific property to the onboarding events:

- Event: onboarding_step_completed where step_name = 'first_message'
- Property: ab_variant — values: 'control' (blank text box) or 'treatment' (3 suggested prompts shown)
- Goal metric: % of users who complete 'first_message' step within 10 minutes of account creation

6. Data Model & SQL Schema

The following schema defines the database tables that will be created to support SQL analysis. This mirrors the structure of a real product analytics data warehouse.

6.1 Core Tables

users

One row per registered user. Contains demographic and acquisition data.

- user_id (PK), product, signup_date, plan_type, country, platform, signup_source, ab_variant

events

One row per tracked event. The core fact table — every user action is a row here.

- event_id (PK), user_id (FK), event_name, event_timestamp, session_id, properties (JSONB)

sessions

One row per session. Aggregated from events — useful for session-level analysis.

- session_id (PK), user_id (FK), session_start, session_end, duration_seconds, message_count, features_used

subscriptions

One row per subscription change. Tracks all plan upgrades and downgrades.

- subscription_id (PK), user_id (FK), from_plan, to_plan, change_date, revenue_usd, churn_date

6.2 Key Relationships

- users → events: one-to-many (one user, many events)
- users → sessions: one-to-many (one user, many sessions)
- sessions → events: one-to-many (one session, many events)
- users → subscriptions: one-to-many (one user, possibly multiple plan changes)

7. Analytical Plan

This section maps out exactly what analyses will be performed, what SQL patterns will be used, and what decisions each analysis will inform.

7.1 Funnel Analysis

We will build a full onboarding funnel from sign-up to first meaningful session using SQL window functions and Mixpanel's Funnel report. The funnel steps are:

1. account_created — user registers
2. onboarding_step_completed (step: profile_setup) — user fills in basic info
3. onboarding_step_completed (step: first_message) — user sends their first query
4. session_started (second session) — user returns for a second session
5. WAU-5 threshold reached — user is classified as 'activated'

SQL technique: CTEs with window functions (LAG, LEAD) and conditional aggregation to compute conversion rates between each step.

7.2 Retention & Cohort Analysis

We will compute classic N-day retention curves using cohort analysis. Users are grouped by their signup week (the cohort) and we measure what percentage returned on Day 1, 3, 7, 14, 21, and 30.

SQL technique: Self-join on the events table with date arithmetic. DATEDIFF between first session date and subsequent session dates. Results will be visualised as a retention heatmap in both Mixpanel and Looker.

7.3 A/B Test Analysis

The 'Suggested Prompts' experiment tests whether showing 3 suggested prompts on the blank message input (treatment) improves Day-1 activation rate compared to a blank input (control).

- Null Hypothesis (H_0): Suggested prompts have no effect on Day-1 activation rate
- Alternative Hypothesis (H_1): Suggested prompts increase Day-1 activation rate
- Statistical test: Two-proportion z-test (comparing conversion rates between groups)

- Significance level: α = 0.05 (95% confidence interval)
- Minimum detectable effect: 3 percentage point lift in activation rate
- Required sample size per group: ~1,100 users (calculated via power analysis at 80% power)

7.4 Feature Correlation Analysis

We will identify which features — code generation, image input, memory/personas, plugins — are most strongly correlated with 30-day retention. This uses SQL aggregation joined to retention flags.

Important note on methodology: Correlation is not causation. We will acknowledge that users who adopt advanced features may simply be more engaged to begin with (selection bias). Where possible, we will control for tenure and plan type.

8. Dashboard & Reporting Plan

8.1 Mixpanel Dashboards

Three Mixpanel dashboards will be created, each serving a different audience:

- Onboarding Funnel Dashboard — tracks step-by-step conversion in real time; audience: Growth team
- Retention & Engagement Dashboard — weekly retention curves, DAU/MAU, WAU-5 trend; audience: Product team
- A/B Test Results Dashboard — live experiment results with significance indicators; audience: Product + Engineering

8.2 Looker Executive Dashboard

A single Looker dashboard using LookML will serve the executive audience. It will show:

- North Star metric (WAU-5) trend over 90 days with target line
- Revenue metrics: MRR, conversion rate, average revenue per user (ARPU)
- Comparative product performance: all four AI products side by side on key metrics
- Cohort retention heatmap: colour-coded by retention strength
- One-line data story: plain-English automated insight generated from the metrics

9. Success Criteria

This project will be considered analytically complete when the following deliverables are produced and the following questions can be answered with data:

DELIVERABLE 1	SQL query library — 15+ documented queries covering all 5 analytical question areas, committed to GitHub with README explanations.
DELIVERABLE 2	Mixpanel live dashboard — 3 dashboards with funnel, retention, and A/B test reports populated with synthetic data.
DELIVERABLE 3	Looker executive dashboard — LookML model and Explore with 6 visualisations served from BigQuery.
DELIVERABLE 4	Insights deck — 10-slide presentation following the Pyramid Principle narrative structure, suitable for a VP of Product audience.
DELIVERABLE 5	Written case study — 1,500-word Notion/Medium article explaining methodology, findings, and limitations.

10. Risks & Limitations

Being transparent about limitations is a hallmark of a rigorous analyst. The following must be acknowledged in all public-facing deliverables:

- **All data is generated using Python and does not represent real user behavior. Patterns are illustrative, not empirically validated.** Synthetic data
- **Correlations observed between features and retention do not imply causation. A true causal study would require randomised experimentation.** No causal inference
- **Cohort analysis may understate churn if the data generation script over-represents retained users. This will be explicitly noted.** Survivorship bias
- **Real product comparisons would be confounded by differences in user acquisition channels, pricing, and geography across the four products.** Platform differences

11. Project Timeline

Phase	Week	Deliverables	Tools
1	1	PRD writing, problem framing, metric definition	Google Docs
2	1–2	Synthetic data generation, Mixpanel event ingestion	Python, Mixpanel, CSV
3	2–3	SQL analysis: funnel, cohort, A/B test, feature correlation	PostgreSQL, BigQuery
4	3	Mixpanel dashboard build: 3 dashboards	Mixpanel
5	4	Looker dashboard + LookML model	Looker, BigQuery, LookML

6	4–5	Insights deck and written analysis	Google Slides, Notion
7	5	Portfolio packaging: GitHub, LinkedIn, case study	GitHub, LinkedIn, Notion

12. Glossary

Key terms used throughout this document:

- DAU / MAU — Daily / Monthly Active Users. Measures how many unique users engage with the product per day or month.
- Cohort — A group of users who share a common characteristic, typically their sign-up week. Cohort analysis tracks how behavior evolves over time.
- Funnel — A sequence of steps in a user journey. Funnel analysis measures the conversion rate from one step to the next.
- Retention — The percentage of users who return to the product after a given number of days from their first visit.
- Churn — When a user stops using the product. Often defined as no activity for 30+ days.
- A/B Test — A controlled experiment where two variants (control vs. treatment) are served to randomly split user groups to measure the effect of a change.
- p-value — The probability of observing a result at least as extreme as the one measured, assuming the null hypothesis is true. $p < 0.05$ is typically considered statistically significant.
- North Star Metric — A single metric that best captures the core value the product delivers to users. All other metrics should move in service of the North Star.
- LookML — Looker's data modelling language. Defines the logic for how raw database tables are queried and presented in Looker dashboards.
- WAU-5 — Weekly Active Users who complete 5 or more conversation turns in a given week. This project's North Star Metric.

E N D O F D O C U M E N T

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